

BUSINESS CASE

Problem Statement

Traditional access systems:

- Cards → can be stolen
- PINs → can be shared
- Manual security → inefficient

Need: **Secure, automated, contactless identity verification**

Proposed Solution

A **cloud-native biometric access control system** using AWS:

- Fully serverless
- AI-driven (face recognition)
- Real-time authentication
- Highly scalable

Business Value

1. Security Enhancement

- Eliminates impersonation
- Biometric identity = unique
- No password/card theft

2. Operational Efficiency

- No manual verification
- Instant access decisions (<1 sec)
- Reduced security staff overhead

3. Cost Optimisation

- Serverless → pay per request
- No infrastructure maintenance
- DynamoDB on-demand scaling

4. Scalability

- Supports:
 - 10 → 10,000 employees seamlessly
- Auto-scales with usage

5. User Experience

- Contactless entry
- Fast and seamless
- Modern digital workplace

Risks & Mitigation

Risk	Mitigation
False positives	Adjust similarity threshold
Lighting issues	Enforce capture standards
Privacy concerns	Encrypt + IAM + policies
Latency	Use regional services (eu-west-1)

ROI Justification

- Reduced manpower costs
- Improved security compliance
- Faster onboarding/offboarding
- Scalable across multiple office locations

Break-even achieved quickly for medium-large enterprises

5. Final Assessment

We have implemented:

- Event-driven architecture
- Serverless microservices
- AI/ML integration
- Secure IAM model
- Real-time API system

- Production-grade monitoring

This is **enterprise-grade architecture**